

CSA05-DATABASE MANAGEMENT SYSTEMS

LAB OBSERVATION

EXPERIMENT:19

AIM:

To implement trigger in MySQL.

Problem:

Program 1 :

Create a trigger in MySQL to log the changes of the STUDENT table with fields ID, Name and Email. Also create a new table named EMPLOYEES_AUDIT to keep the changes of the employee table. Create a BEFORE UPDATE trigger that is invoked before a change is made to the employees table.

SQL QUERIES:

```
mysql> CREATE TABLE STUDENT (ID INT PRIMARY KEY, Name VARCHAR(30), Email VARCHAR(60));
Query OK, 0 rows affected (0.04 sec)
mysql> CREATE TABLE EMPLOYEES (ID INT PRIMARY KEY, Name VARCHAR(30), Email VARCHAR(60));
Query OK, 0 rows affected (0.03 sec)
mysql> CREATE TABLE EMPLOYEES_AUDIT (ID INT, Name VARCHAR(30), Email VARCHAR(60),
-> action VARCHAR(20), changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
Query OK, 0 rows affected (0.03 sec)
mysql> DELIMITER //
mysql> CREATE TRIGGER student_log
-> AFTER UPDATE ON STUDENT
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO EMPLOYEES_AUDIT(ID,Name,Email,action)
-> VALUES(OLD.ID,OLD.Name,OLD.Email,'STUDENT UPDATE');
-> END //
Query OK, 0 rows affected (0.01 sec)
mysql> CREATE TRIGGER employees_before_update
-> BEFORE UPDATE ON EMPLOYEES
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO EMPLOYEES_AUDIT(ID,Name,Email,action)
-> VALUES(OLD.ID,OLD.Name,OLD.Email,'BEFORE UPDATE');
-> END //
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
mysql> INSERT INTO EMPLOYEES VALUES (101,'Arun','arun@gmail.com');
Query OK, 1 row affected (0.01 sec)
mysql> UPDATE EMPLOYEES SET Email='arun_new@gmail.com' WHERE ID=101;
Query OK, 1 row affected (0.01 sec)
mysql> SELECT ID,Name,Email,action FROM EMPLOYEES_AUDIT;
+-----+-----+-----+-----+
| ID | Name | Email | action |
+-----+-----+-----+-----+
| 101 | Arun | arun@gmail.com | BEFORE UPDATE |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
mysql>
```

RESULT:

Thus the trigger is executed successfully.